Figure 10 illustrates the software architecture for the Lx FMRI mode operation of the system. In this mode, the operator prescribes the imaging control parameters for series to be acquired, saves the prescription, and then executes the acquisition. This opens the FMRI acquisition UI (BWRT). This user interface allows users to view and save processed images into database.

Figure 10 Host Software Architectural Breakdown – Lx FMRI

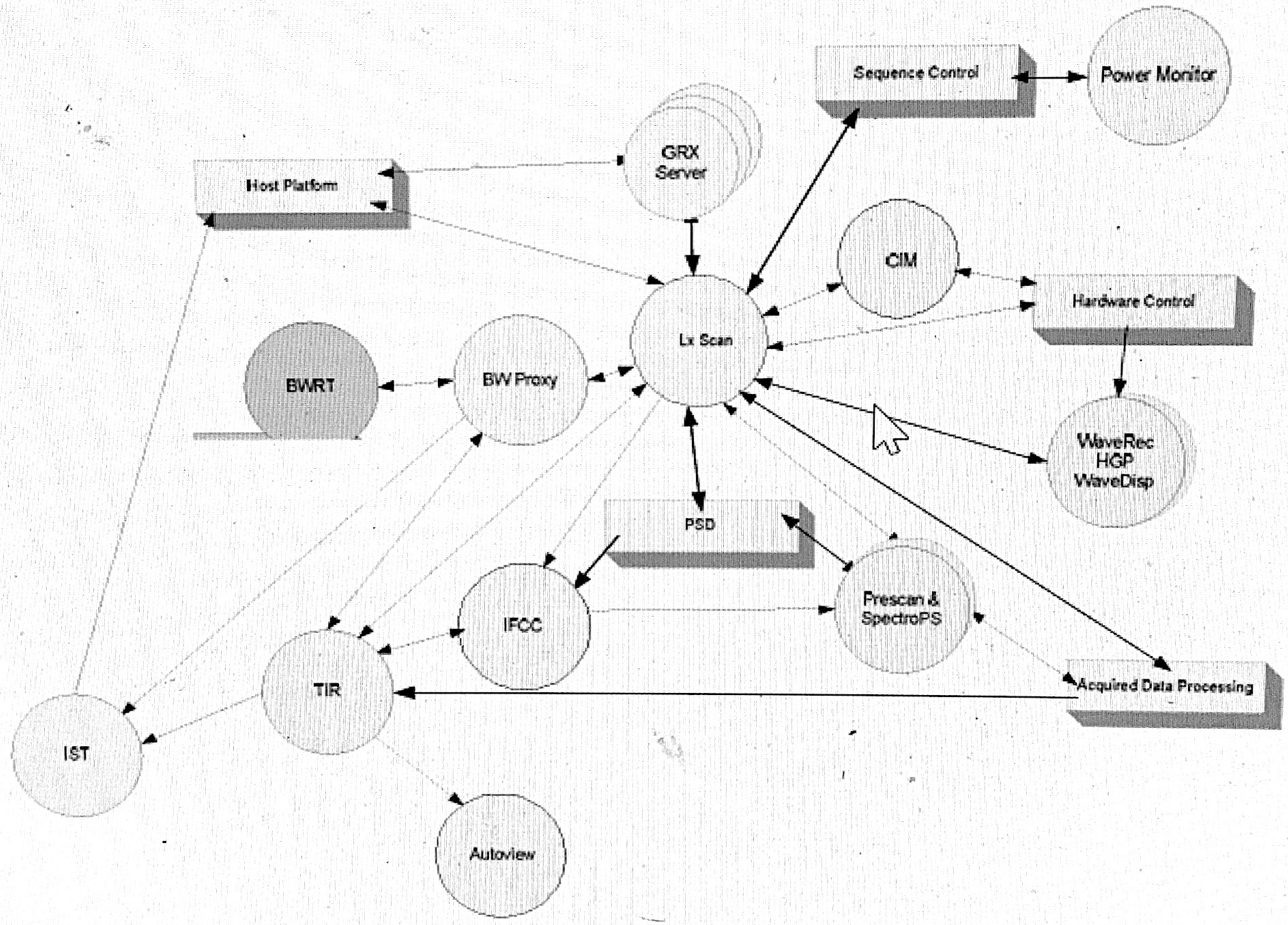
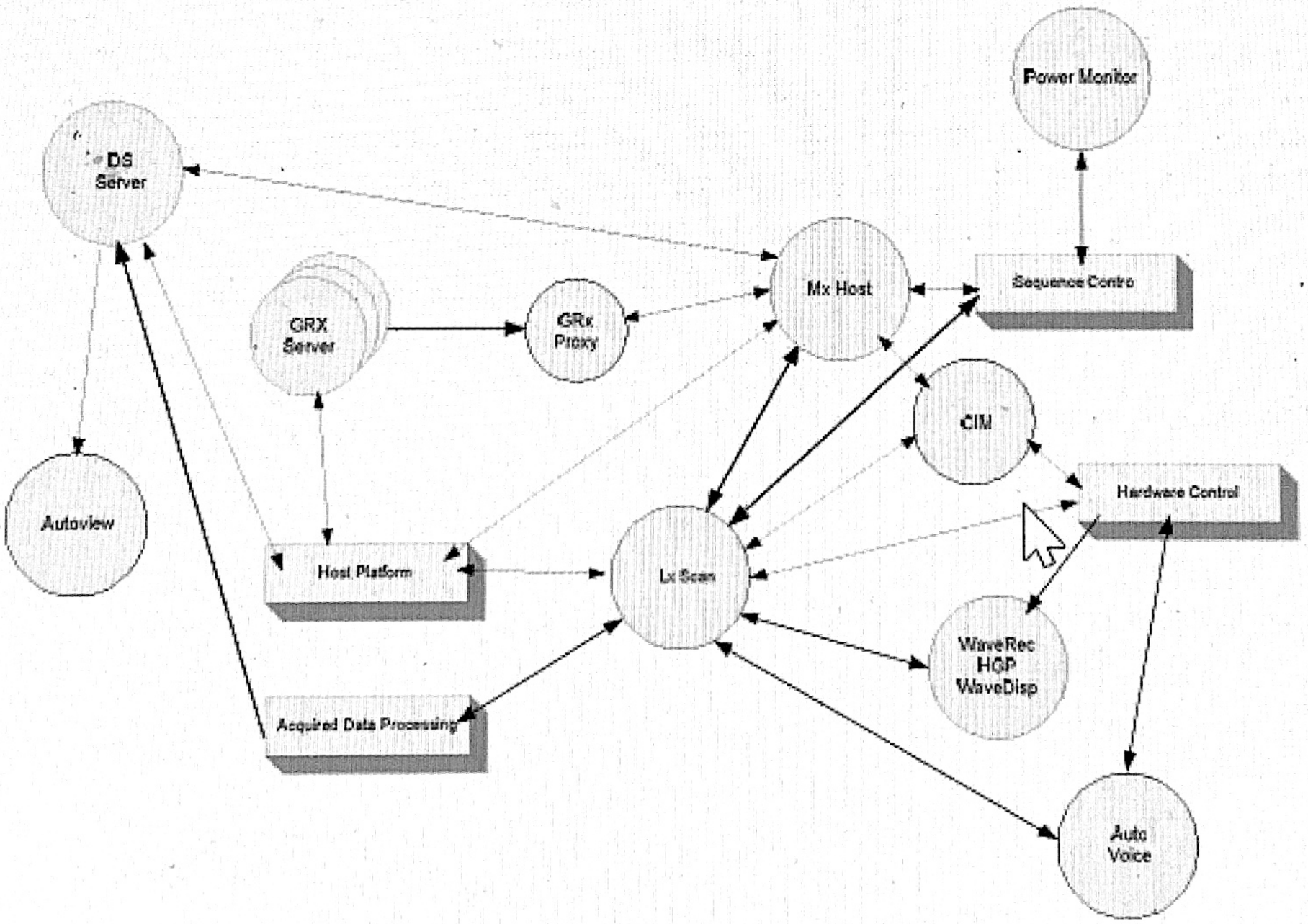


Figure 11 illustrates the software architecture for the Mx mode operation of the system. This mode allows both batch & real-time scanning. In batch mode, the operator prescribes the imaging control parameters for series to be acquired, saves the prescription, and then executes the acquisition. While the acquisition is in progress the operator is free to prescribe additional series or perform post processing or image viewing operations. The system is capable of automatically acquiring (sequentially) a set of prescribed series when running with auto-scan enabled. In real-time mode it brings up acquisition UI that displays acquired images and allows parameter change on fly and acquire images. It provides facility to switch to high resolution mode, complete acquisition and return to real-time mode.

Figure 11 Host Software Architectural Breakdown - Mx



Released

Table 2, below, shows the association between the components depicted in Figure 8 through Figure 11 and the high level functional groups shown in Figure 2.

Table 2 Functional Grouping of Host Software Components

Component	Functional Grouping	Notes
Lx Scan	Application Launch, Rx UI, Geometry, Acquisition UI, State Machine, Hardware interface	Interfaces with host platform in managing start /end exam and task list. Stores and manages data for all applications. Supports Lx applications for all stages. Provides interface to start Mx applications. Manages hardware events and provides needed data to hardware.
Mx JVM	Application Launch, Rx UI, Geometry, Acquisition UI, State Machine, Hardware interface	Supports Mx applications
CIM	Coil Information Manager	The coil data management facility comprises the software that controls access to the coil database, process coil codes, provides clinical coil and coil data to coil data consumers as well as compute the settings necessary to operate the system using connected coils.
SVAT, EXSI	External Interface	Control interface for programmatic control of prescription and scanning.
Prescan & SpectroPS	Acquisition UI	Provides user interface to the prescan capability for both imaging and spectroscopy.
GRX Server (2 sets)	Graphic Rx	Controls image canvas for the graphic Rx capability
GRx Proxy	Graphic Rx	Communication interface between C++ GRX server and Mx JVM.
Autoview	Acquisition UI	Image display of recently acquired images
DS Server	Image Receive Chain	Mx side of image receive chain
BW Proxy	Acquisition UI	Communication interface between GEHC MR software and BrainWave RT application.
RXM	Rx UI	It is prescription UI for FMRI applications
PXE	Acquisition UI	It is paradigm execution engine
BW RT	Acquisition UI	It displays & processes FMRI images.
IFCC	Image Receive Chain	Image File Creation and Cleanup
TIR	Image Receive Chain	Lx side of image receive chain for batch and fMRI scans.
IST	Image Receive Chain	Image save interface from TIR to database for fMRI scans.

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Component	Functional Grouping	Notes
APP	Image Receive Chain	Automated post processing control
Auto Post Processing	Image Receive Chain	Automated post processing image generation
RTIA	Acquisition UI	Lx interactive real-time control
Display Server	Acquisition UI	Lx interactive real-time image display
RIR	Image Receive Chain	Lx image receive chain for RTIA
RTPSI	Acquisition UI	Lx real-time parameter communication interface
ILT	External Interface	Image file interface for external applications
WaveRec & WaveDisp	Acquisition UI	Gating waveform display
NAV	Acquisition UI	Navigator display and control
Host Power Monitor	Acquisition UI	Displays SAR and energy absorbed by patient
Express Spine Plugin	External Interface	Used by core viewer to run auto spine algorithm
One Touch Spine Library	External Interface	Algorithm used to get vertebrae labeling for spine images
HOS	Acquisition UI	Graphical Prescription for shim
Tie	External Interface	Emulates MGD
Localizer Viewer	Acquisition UI	Viewing screen save images

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The following paragraphs further describe each of these components.

3.1.1 Lx Scan

This process interfaces with host platform to support start exam, task list management, protocol management and end exam. It supports complete life cycle of Lx applications that includes Rx UI control, acquisition controls and state machine. It interfaces with PSD subsystem in controlling different parameters in RX UI. It interfaces with Graphic Rx in sending and receiving graphical data. It interfaces with IFCC for communicating prescription information. It interacts with firmware sub-system in managing hardware events. It interfaces with RECON sub-system for controlling scanner state machine. It interfaces with RTIA to support real-time applications. It also provides mechanism to start Mx applications.

3.1.2 Mx JVM

The Mx JVM is the main host process for the Mx platform. It is responsible to host the Mx application, to
provide the user interface services for the Mx platform and applications, and to provide the Mx state machine to manage the state transitions for the Mx applications. It communicates with the hardware subsystems to define setup and run the acquisition applications. It runs under the direction of the Lx scan process which hands off system control to Mx to run Mx applications. It obtains exam level parameters from the Lx scan process and communicates changes back to the Lx scan. It hosts the RX UI for Lx
applications.

3.1.3 CIM

The coil data management facility comprises the software that controls access to the coil database, process coil codes, provides coil data to coil data consumers as well as compute the settings necessary to operate the system using connected coils. For the GEM coil set, the coil data management manages the data necessary to locate floating coils and to generate the data necessary to drive the automatic coil selection and automatic calibration series generation. Additionally, the JSTD curve data associated with the SAR calculations are managed within the coil database. For dynamic coil configuration, dynamic coil mode is generated from selected REGs and element data by the autoCoil selection algorithm.

3.1.4 LAIS/ATP/SVAT/ILT

The LAIS/ATP/SVAT processes provide the programmatic interface to the scanner. The LAIS is used
for remote control and primarily used by focused ultrasound. The ATP provides the scripting capability,
primarily used by service tools. The SVAT is the command director, which receives the command

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stream from LAIS or ATP and forwards the commands to the correct process on the host. The ILT (image listener task) is the Image file interface to provide a near real-time image data stream for external applications.

3.1.5 Prescan & SpectroPS

The host prescan process is responsible to provide the user interface and controls to set the system tuning parameters relevant to imaging acquisitions. This parameter set includes center frequency, transmit gain, gradient shim values, and digital and analog receive gains. It also provides user interface to modify phase delay and amplitude attenuation for optimized dual drive mode. For dynamic coil configuration, it generates pre-processed reference data for later consumption by the auto coil selection algorithm.

The spectro prescan process is responsible to provide the user interface and controls to set the system tuning parameters relevant to spectroscopy acquisitions. This parameter set includes center frequency, transmit gain, gradient shim values, digital and analog receive gains, zero order and first order phase and line broadening.

3.1.6 GRX Server (2 sets)

The GRx Server process shall manage the image display and user interactions in a single graphic Rx view port. There are three instances of the same process, one per graphic Rx view port. The GRx Server process interaction is on the image canvas via the mouse (and also accepts keyboard short cuts). The three GRx Server processes communicate between each other to maintain a consistent view of the current prescription across the displayed localizer images.

3.1.7 GRx Proxy

The GRx Proxy provides the communication interface between C++ GRX server and Mx JVM. It communicates events to GRX Server using x-atoms and data using shared memory. It connects to Mx JVM using sockets.

3.1.8 Autoview

Autoview provides the immediate display of acquired images. It has two modes of operation. In one mode it automatically updates the acquired images. In another mode it updates only after user request.

3.1.9 DS Server

The DS Server is responsible to receive static image parameters from the Mx application (running on the Mx JVM), to receive the image data stream plus dynamic image parameters from the real time subsystems, and to join that data into DICOM images to be handed off to the Image Database for storage. It also sends the images to the autoview component for immediate display for the operator.

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3.1.10 BW Proxy

The BWProxy application launches the brainwave real-time processing application at the beginning of a fMRI scan, which utilizes the brainwave real-time processing application. While in research mode the user can disable real-time processing. The BWProxy application provides the interface between the image stream and the brainwave real-time processing and display application as well as the interface between scan and the brainwave real-time processing application to pass scan state information.

3.1.11 RXM

RXM is used to prescribe a fMRI series. This is launched from prescription screen. It is used to input/select the paradigm information, which needs to be scanned. It is also used to connect PXE for creating new software paradigms and to see the training paradigms.

3.1.12 BWRT

BrainWaveRT is a real-time viewer for the fMRI study. This is launched once the fMRI scan is prepped. It is used to see the real-time images from the functional scanning. It is a brain mapping software to record the acquisition data. It integrates the sensor device with the software and is used to start functional brain acquisition and the real-time visualization of brain maps. User scan change the background , Z value and other parameters from the Brainwave RT screen.

3.1.13 PXE

PXE is the Paradigm eXecution Engine responsible for playing a visual or audio stimulus to the patient. When the operator selects a hardware paradigm, PXE is loaded with the stimulus material. PXE requires a third display device to be available on the scanner as well as an input trigger sent by the scanner at the start of the scan.

3.1.14 IFCC

The IFCC process provides the image file creation and cleanup processing for the Lx platform. This process provides space management for the acquisition subsystem, allocating image file space prior to the beginning of each acquisition. The IFCC process is responsible to create image headers and communicate reconstruction parameters to recon. At the end of acquisition this process is responsible to de-allocate all unused space from the acquisition.

3.1.15 TIR

The TIR process manages the image receive and database installation for the Lx platform, except for RTIA images. The TIR process is responsible to receive the stream of images from the reconstruction subsystem and pass the images to the image database. For fMRI acquisitions the IST process acts as the intermediary between TIR and the database to isolate database install

performance from real-time image transport. TIR also sends the images on to the autoview via a shared memory circular buffer. The TIR process reports back to the scan and IFCC processes the scan status once recon has completed and all reconstructed images have been either installed into the database or discarded due to error conditions.

3.1.16 IST

The IST (Image Save Task) process receives fMRI images from TIR, installs the images into the database and provides installation status back to TIR. The IST process is only active for fMRI real-time acquisitions. It provides performance buffering between TIR and the Database image install processing

3.1.17 APP

The APP (automated post processing) application manages auto-subtract on acquired images. The APP process receives image availability notification from TIR along with the requested processing application ID and launches the automated post processing once sufficient images are available to complete processing.

3.1.18 Auto Post Processing

The automated post processing processes perform the automated post processing including post processed image generation and installation.

3.1.19 RTIA

The RTIA JVM provides the main host side controls for the real-time platform. It is enabled by the Lx
Scan process when a real-time capable PSD is started in real-time mode. This process communicates new parameter values to the PSD via RTPSI (real-time pulse sequence interface), receives updated spatial location information from the display server and hosts the display canvas used by the displayserver. When in review mode the RTIA JVM provides the user controls to drive the display server to
display images selected by the user and to store user selected images into the image database. The RTIA process communicates with the Lx Scan process to pass start/end locations for subsequent acquisitions as well as to hand back system control to Lx Scan once the real-time session has been closed.

3.1.20 Display Server

The Display Server is responsible to display images received from RIR. It provides on screen controls for user, to define new scan plane locations and store images into database. It also provides the
review capability to display images that are in the shared memory image ring buffer for review purposes by the user.

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3.1.21 RIR

The RIR (Real-time Image Receive) process receives the image stream from the reconstruction subsystem, transforms this data into DICOM images and passes the images on to the display server.

3.1.22 RTPSI

The RTPSI (Real Time Tps Interface) process is the Lx real-time parameter communication interface for RTIA. It gets initial parameter values from AGP and sends them to RTIA JVM for UI update. It communicates user request on RTIA UI to AGP. It also allows user to start/stop/pause scanner.

3.1.23 NAV

The NAV (Navigator) process provides the navigator display and control capability for acquisitions using the navigator capability. It is enabled when scan operations is performed using navigator imaging option.

3.1.24 HGP, WaveRec & WaveDisp

The HGP (Host Gating Process) process provides the user interface to enable and control the cardiac and respiratory gating services of the scanner. The Wave Rec process is responsible to receive the waveform and time/view count data from the hardware control. It makes this data available in shared memory. The Wave Disp process is responsible to provide the visual display of the cardiac and respiratory waveform data received by Wave Rec

3.1.25 In Room Display

In Room Display provides interface for the user to view Scan Time, Table position, Gating waveforms, Patient Information and connected coils from magnet room. It also allows the user to change patient position, patient weight, gating controls as well as the auto start option.

3.1.26 Auto Voice

Auto voice provides the capability to play back pre-recorded and stored voice messages from the intercom system at the operator console to the patient using the speakers in the bore of the magnet. When the Auto voice mode is ON, voice generated through the software functions will instruct the patient to hold their breath during scanning and will instruct the patient to resume breathing after the scan has completed. The User will also be able to create and modify existing Auto voice messages using the intercom microphone on the operator console. These messages will be recorded through the Auto voice Manager Record and Play function.

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3.1.27 Power Monitor

This process is responsible for displaying total energy absorbed by patient, SAR limits and scan room temperature. It gets this information from hardware control sub-system and updates the user interface every three seconds. It also computes time left to reach the limit and display it on SAR window along with the total accumulated specific absorbed energy. If the time left is less than equal to 10 min SAR window pops up with an option to override scan stop.

3.1.28 High Order Shim

High Order Shim is used for optimizing the magnetic field homogeneity on a patient-by-patient basis when the B0 uniformity of significantly better than 1 ppm is desired. High Order Shim ROI selection window appears after Prep has been registered. It is loaded with default images. It provides slider bars to select desired images. It supports both rectangular and elliptical ROIs. After positioning the ROI over the anatomy of interest, the optimum values for improving the magnet homogeneity within the specified ROI volume are calculated.

3.1.29 TPS/IPG Emulator (TIE)

This process is used to support the "show system" mode. It simulates real time sub-systems allowing host computer to operate independently. It registers as receiver for all communications to acquired data processing and hardware control. It also sends response to host software and platform components for their requests. This process is started only when system is in "show" mode.

3.1.30 Localizer Viewer

The Localizer Viewer is a single viewport viewer intended to simplify viewing and filming workflow with respect to GRx screen save images. It displays the screen save images (saved by the GRx screen save feature) corresponding to the series selected in the Browser/WFM inline viewer. It supports viewer functionalities like W/L, W/W, zoom, pan, annotation configuration (whole, partial, none & large font support). It also supports filming through drag & drop and through shortcut F1 key.

3.1.31 Express Spine Plugin

The Auto Spine labeling tool will be a plugin into the viewer application to keep the module independent and easily enabled/disabled on the system without having to change any viewer code. This enables reusability of the existing viewer application and its tools required for this feature, like point markers, labeling (user annotation), screen save, filming, etc. The user does not have to switch between applications for this functionality

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3.1.32 One Touch Spine Library

This is algorithm used to get vertebrae labeling for spine images.

3.1.33 External Scanner Interface (EXSI)

EXSI provide the programmatic interface to the scanner. It is used for remote control and primarily used by focused ultrasound. It receives the command stream from clients and forwards the commands to the correct process. It is standardized, published, extensible interface. It does not change from release-to-release, or for each OS change. It is scalable interface and supports multiple concurrent clients.

3.1.34 Dual Drive

The purpose of the Dual Drive is to improve image homogeneity. This is accomplished by developing an architecture that has the ability to control the amplitude and phase relationship between two separate drive points on the system body coil.

3.2 Risk Assignments

Risk requirements for software flow down from the RAC [5]. Safety requirements are identified and verified appropriately.

The components within the host software subsystem associated with Class C risks have been segregated using data checksum redundancy as provided by the host platform for safety risk associated with RF Power, Image Flip and patient identification.
The remaining host software components were not designed to provide clear segregation nor redundancy to address the remaining class B software risk control. Many of the components in the current architecture are legacy components and were developed before GEHC_GQP_10.04 Software Development Lifecycle Procedure was effective (June 5, 2009) and is thus considered legacy code according to the MR SDLC WI (DOC0593175). These software components are now considered class B and no further segregation has been performed

Software requirements are flowed down from the Product System level requirements. These requirements flow down to the respective software items appropriately.

3.3 Internal Software Interfaces

This section provides a high level description of the data, event, and command interfaces between the individual components making up the Host Software Subsystem. Table 3 below shows the communication associations between components. These links will be further described in the remaining subsections within this section. Each of these subsections will define the communications originating from that component to other components. If a reply is expected, it will be noted. The reply

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will be considered to "originate" from the other endpoint and will be covered in that section. The following table shows the format and describes the content.

To	Type	Description	Interface
Component communication is being sent to (or replied to)	Request – if this is a request (reply is expected) Reply – if this is a reply to a request Event – if this is signaling an event, no reply expected. Send – if this is a send w/o an expected reply FILE- W File Write FILE-R File Read	Description of the data being transported.	Communication interface (e.g. MrMail, Corba ...)

Table 3 Component to Component Communications

Intercomponent communications: FROM \ TO	Lx Scan	Mx JVM	CIM	SVAT	EXSI	Prescan & SpectroPS	GRX Server (set 1)	GRX Server (set2)	GRx Proxy	Autoview	DS Server	BW Proxy	IFCC	TIR	IST	APP	Auto Post Processing	RTIA	Display Server	RIR	RTPSI	ILT	NAV	WavDisp	HGP	AutoVoice	Acquired Data Processing	Host Platform	Hardware Control	Sequence Control	Pulse Sequence	External Scanner Control Device	BWRT
Lx Scan		x	x	x	x	x	x					x	x	x				x						x	x	x	x	x	x	x	x		
Mx JVM	x		x						x		x																x	x	x	x			
CIM	x	x																									x	x	x	x			
SVAT	x				x	x							x																x				
Prescan & SpectroPS	x			x	x								x															x					
EXSI																												x					
GRX Server (set 1)	x																											x					
GRX Server (set 2)									x																								
GRx Proxy		x						x																				x					
Autoview								x																				x					
DS Server		x								x																							
BW Proxy	x													x	x												x						x
IFCC	x					x								x																			
TIR	x									x		x	x			x				x							x						
IST												x				x					x						x	x					
APP													x															x					
Auto Post Processing														x														x					
RTIA	x														x																		
Display Server																			x	x	x							x					
RIR					x													x		x													
RTPSI					x								x					x	x			x					x						
ILT					x													x									x						
NAV												x		x						x							x						
WavDisp																																	
HGP																																	
AutoVoice																									x			x	x				
Acquired Data Processing																													x				
Host Platform	x	x			x	x	x	x			x		x	x						x	x		x						x				
Hardware Control	x		x											x	x		x		x					x									
Sequence Control	x	x																						x		x							
Pulse Sequence	x																																
External Scanner Control Device				x	x								x																				
BWRT												x										x											

3.3.1 Lx Scan Communications

This process is active all the time. The following communications originate from the Lx Scan component:

To	Type	Description	Interface
Host Platform	Send	Start Exam status, Scan status, Scan time Data set information, Task information, scan options, research options, dual drive information	GEOCOM
Mx JVM	Send	UI Information for Lx applications	CORBA
Mx JVM	Send	Patient Information, Series information, Landmark Information, PPS information, Calib information, Smart Prescan information, Driver information, Entry Point Table information, Coil Used information, Shim Vol Information, Contrast information, Auto Scan information, Auto Voice information, GRX information	CORBA
Mx JVM	Send	Start/End Exam, Show/Hide UI Show/Hide GRX, Launch /Save Protocol, Save/Download/Scan, TPS Reset Start/Stop, Landmark Changed, Desktop changed, Coil Changed, Execute SVAT command, Scan mode changed, SAR/dbdt mode changed	CORBA
CIM	Request	Coil Information	MrMail
CIM	Event	Start/End Exam	MrMail
CIM	Request	Auto Coil Information, Auto Calib Information	GEOCOM
SVAT	Reply	Status of the command and information	MrMail
Prescan	Send	Start Prescan event with Rx information (including sequence number)	MrMail
Prescan	Event	To bring prescan UI to the front	MrMail
Spectro	Event	Start /Stop spectro prescan, Landmark /CV download, End Exam	MrMail
NAV	Send	Scan start information/ End Exam	MrMail
Grx Server	Event	User Operations	X-Atoms

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Grx Server	Send	Graphical Rx Information, Coil Extents	Shared memory
BWProxy	Send	RT UI Control, Rx information	MrMail
BWProxy	Event	Scan Event/Shutdown	MrMail
IFCC	Send	Rx information	MrMail
IFCC	Event	Stop/Abort/Cleanup/End Exam	MrMail
TIR	Send	TDEL update for dynaplan	MrMail
TIR	Send	FUS Information, Coil Data	MrMail
RIR	Send	Template header path	MrMail
RTIA	Event	Wakeup	Corba
Wave Display	Send	Scan/Wait time for dynaplan	MrMail
HGP	Event	Wakeup	MrMail
Auto Voice	Event	Start, TPS Reset, Start Exam, End Exam	Corba
IRM	Publish	Patient Table Status and Position	GEOCOM
Host Platform	Request	Query DataBase	Query
Host Platform	Send	PPS Information	MrMail
Host Platform	Send	Diag information	Query
Server Prescan	Send	Filters/Attention	MrMail
Hardware Control	Send	Option key information, Table ISO, Scan button mode for dynaplan, Auto Start /Auto Table /Auto Voice settings, Advance to Scan, Entry Point Table, Patient Entry/Position	MrMail
Hardware Control	Query	Table entry, Door usage status, Landmark location	MrMail
Hardware Control	Send/Query	Gating selections	MrMail
Acquired Data Processing	Send	BAM check, Ready Packet, Scan start, Scan Stop packet	MrMail
Sequence Control	Send	RSPs	MrMail
Pulse Sequence	Send	Rx Information	Message Q
Pulse Sequence	Query	Filters, UI Information, Load/Download CVs	Message Q
EXS	Reply	Status of the command and information	GEOCOM

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3.3.2 Mx JVM Communications

This process is active for Mx applications. The following communications originate from the Mx JVM component:

To	Type	Description	Interface
SCN	Send	User actions on Rx UI for Lx applications	CORBA
SCN	Send	Series information, GRX information, Landmark information, smart prescan information, Contrast information	CORBA
SCN	Send	Mx Ready, app launched, download done, scan started, scan stopped, first image installed, start session done, session finished, mx svat done, launch protocol done, save protocol done, save rx done, update series description, set graphic info done	CORBA
CIM	Request	Installed coils and connected coil for displaying coil selection UI Coil information and Hardware settings	GEOCOM
Grx Proxy	Send/Receive	Grx Events and Information. Coil Extents.	Socket
Host Platform	Send	Scan time, Scan status	GEOCOM
DsServer	Send	Representation of the static part of the image header and the connections to receive the dynamic parts of the header	Tags
Acquired Data Processing	Send	Hardware independent version of processing instructions to transform (usually) raw data to image data.	Tags
Hardware Control	Send	Representation of the instructions to initialize and control hardware. Also may control the overall application specific scan state for the real-time servers.	Tags

3.3.3 CIM Communications

This process is active for all the time. The following communications originate from the CIM component:

To	Type	Description	Interface
SCN	Event	Initialization	MrMail
Hardware Control	Event	Initialization	MrMail
SCN	Send	Coil Connect Events Coil Code	MrMail
Mx JVM	Send	Coil Connect/Disconnect Information	GEOCOM

3.3.4 SVAT Communications

This process is active when svat commands are used. The following communications originate from the SVAT component:

To	Type	Description	Interface
Host Platform	Send	Command with arguments	GEOCOM
SCN	Send	Command with arguments	MrMail
IFCC	Send	Command with arguments	MrMail
HGP	Send	Command with arguments	MrMail
Prescan	Send	Command with arguments	MrMail
EXSI	Send/Recieve	Command with argument, status and information	CUP

3.3.5 Prescan & SpectroPS Communications

This process is active during Lx prescan and spectro prescan. The following communications originate from the Prescan and Spectro prescan components:

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To	Type	Description	Interface
SCN	Send	Prescan (Both auto and manual) done event with status and information	MrMail
SCN	Send	Spectro Prescan done event with status and information	MrMail
SCN	Request	To display RSPs	MrMail
SCN	Send	Update prescan information to Mx	MrMail
Acquired Data Processing	Send	Rx Information, User modification on manual prescan UI and spectro UI	MrMail
SVAT	Reply	Status of SVAT commands	MrMail
Host Platform	Send	Dual Drive Parameters	GEOCOM
Prescan	Send	User operation on spectro UI	MrMail

3.3.6 GRX Server (set 1) Communications

This process is active when Lx applications are setup. The following communications originate from the GRX Server component:

To	Type	Description	Interface
SCN	Event	User operation	X-Atom
SCN	Send	Graphical Prescription Data	Shared Memory
Other Grx Servers	Event	User operation	X-Atom
Other Grx Servers	Send	Graphical Prescription Data	Shared Memory
Host Platform	Query	Localizer images	AIM

3.3.7 GRX Server (set 2) Communications

This process is active when Mx applications are setup. The following communications originate from the GRX Server component:

To	Type	Description	Interface
GRx Proxy	Event	User Operation	X-Atom
GRx Proxy	Send	Graphical Prescription Data	Shared Memory
Other Grx servers	Event	User operation	X-Atom

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Other Grx Servers	Send	Graphical Prescription Data	Shared Memory
Host Platform	Query	Localizer images	AIM

3.3.8 GRx Proxy Communications

This process is active when Mx applications are setup. The following communications originate from the GRX Proxy component:

To	Type	Description	Interface
GRxServer	Event	User Operation	X-Atom
GRxServer	Send	Graphical Prescription Data	Shared Memory
Mx JVM	Send	User Operation	Sockets

3.3.9 Autoview Communications

This process is active when images are generated for both Lx and Mx applications (Except for RTIA and MR-Echo). The following communications are received by autoview component:

From	Type	Description	Interface
TIR	Receive	DICOM Image	GVTK Shared Buffer
DsServer	Receive	DICOM Image	GVTK Shared Buffer

3.3.10 DS Server Communications

This process is active during generation of images by Mx applications. The following communications originate from the Ds Server component:

To	Type	Description	Interface
Host Platform	Send	Image	File I/O
Autoview	Event	Image arrived	GVTK Shared Buffer
Mx JVM	Event	Recon Done	Tags

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3.3.11 BW Proxy Communications

This process is active for FMRI applications. The following communications originate from the BW Proxy component:

To	Type	Description	Interface
SCN	Event	Scan Events, RT Ready, FMRI Done	MrMail
IST	Send	Image Path	MrMail
BWRT	Send	DICOM Image	BW Interface

3.3.12 IFCC Communications

This process is active while scanning Lx applications. The following communications originate from the IFCC component:

To	Type	Description	Interface
SCN	Event	Allocation / Dataset/ Cleanup Done	MrMail
TIR	Send	Image Header	File I/O
TIR	Send	Template header information	MrMail
Host Platform	Event	Connection to Image Database	AIM
RIR	Send	Template header Path	MrMail
SVAT	Reply	Status of command	MrMail
Pulse Sequence	Query	Image header & Recon Control Variables	Message Q
DataMine	FILE-W	Rx Information	File I/O

3.3.13 TIR Communications

This process is active while scanning Lx applications. The following communications originate from the TIR component:

To	Type	Description	Interface
SCN	Send	Exam Path/ Recon Done /Install Done	MrMail
Autoview	Event	Image arrived	GVTK Shared Buffer
ILT	Event	Image ready	MrMail
APP	Send	Post Processing Information	MrMail
IFCC	Send	Screen Save status	MrMail

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IST	Send	New Image Acquired	MrMail
IST	Send	Acquisition Complete	MrMail
BwProxy	Send	New Image Acquired	MrMail
BwProxy	Send	Acquisition Complete	MrMail
Host Platform	Send	DICOM Image to insert into Database	TAP

3.3.14 IST Communications

This process is active when images are generated for FMRI applications. The following communications originate from the IST component:

To	Type	Description	Interface
TIR	Event	Image Install	MrMail
Host Platform	Send	DICOM Image to insert into Database	TAP.

3.3.15 APP Communications

This process is active during post processing. The following communications originate from the APP component:

To	Type	Description	Interface
IFCC	Send	Processing done	MrMail
Host Platform	Send	PPS Information	MrMail
Host Platform	Send	DICOM Image to insert into Database	TAP

3.3.16 RTIA Communications

This process is active during Lx-RTIA scanning. The following communications originate from the RTIA JVM component:

To	Type	Description	Interface
Display Server	Event	Display/Save-Image command	Corba
Display Server	Send	Navigation mode/ Rotation/Display modes/ Annotation levels/Play mode/Rx mode/ Parameter selection	Corba

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Display Server	Query	Get pixels for image	Corba
RIR	Event	Start/Stop/Pause pulling images	Corba
RTPSI	Event	Start/Stop/Pause Scanner	Corba
RTPSI	Query	Get/Update RSPs	Corba
SCN	Query	Scanning Locations	Corba
SCN	Send	Scanning Locations	Corba
SCN	Send	Image Path/Save Series/Start Hi Res/ Close Session	Corba

3.3.17 Display Server Communications

This process is active during Lx-RTIA scanning. The following communications originate from the Display Server component:

To	Type	Description	Interface
RTIA	Send	Image Arrived information	Corba
RTIA	Query	Image Details in Buffer	Corba
RTIA	Send	Viewport selection Line/Point location	Corba
RTIA	Send	Save Status	Corba
Host Platform	Send	DICOM Image to insert into Database	TAP

3.3.18 RIR Communications

This process is active during Lx-RTIA scanning. The following communications originate from the RIR component:

To	Type	Description	Interface
Display Server	Send	Image	GVTK Shared Buffer

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To	Type	Description	Interface
Hardware Control	Send	Start/Stop/Pause Scanner	MrMail

3.3.20 ILT Communications

This process is active when image transfer is requested using SVAT command. The following communications originate from the ILT component:

To	Type	Description	Interface
TIR	Event	Start/Stop Sending Images	MrMail

3.3.21 LAIS Communications

This process is active when scanner is control by external device. The following communications originate from the LAIS component:

To	Type	Description	Interface
SVAT	Send	Command	cup

3.3.22 NAV Communications

This process is active during navigator Scanning. The following communications originate from the NAV component:

To	Type	Description	Interface
Acquired Data Processing	Send	Register	MrMail
Acquired Data Processing	Send	Thresh hold/acceptance/coil value	MrMail

3.3.23 HGP; WaveRec & WaveDisp Communications

This process is active when user requests for gating waveforms. The following communications originate from HGP, WaveDisp and WaveRec components:

To	Type	Description	Interface
Host Platform	Query	Service Key Status	MrMail

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Hardware Control	Event	Reset, Record ECG, Gating selections	MrMail
Hardware Control	Event	Gating control	MrMail
Lx Scan	Send	Gating selections	MrMail
SVAT	Reply	Command status	

3.3.24 AutoVoice Communications

This process is active during prescription and scanning when autovoice is enabled. The following communications originate from the AutoVoice JVM component:

To	Type	Description	Interface
SCN	Send	Error Message	CORBA

3.3.25 In Room Monitor

This process is active all the time

To	Type	Description	Interface
CIM	Send	Coil Position Request	GEOCOM
HGP	Send	Gating Selection	GEOCOM

3.3.26 EXSI

This process is active when scanner is controlled by external device.

To	Type	Description	Interface
SCN	Send	Command and arguments	GEOCOM
WFM	Send	Command and arguments	GEOCOM
RIR	Send	Command and arguments	GEOCOM
RTPSI	Send	Command and arguments	GEOCOM